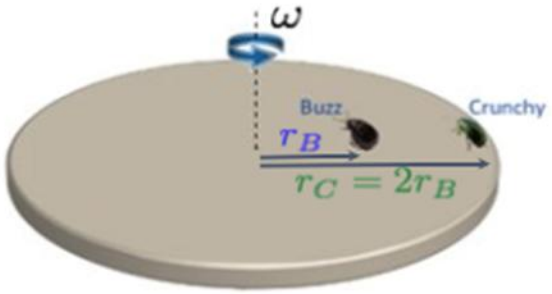
Quiz#9

Section: _____ **Group:** _____

Name(print): _____ **Student ID:** _____

1. Two bugs, Buzz and Crunchy, are standing on a rotating record disk as shown in the diagram below. Which of the following statements is correct?



   A. They have the same angular speed.
   B. They have the same linear speed.
   C. They have the same kinetic energy.
   D. None of the above is correct.

The following questions are based on this situation: A girl is at a carnival riding a Merry-Go-Round that is rotating counterclockwise with an angular velocity of 3.2 rad/s. At a certain moment, the ride begins to slow down with a constant angular deceleration of 0.34 rad/s^2. The distance between the girl and the center of the Merry-Go-Round platform is 1.4 m.

2. What are the girl's linear speed and angular speed after the ride has been slowing down for 3.5 seconds?

3. From the moment the ride begins to slow down, what is the angular displacement of the girl before the Merry-Go-Round comes to a complete stop?

4. Which of the following diagrams correctly show the directions of angular velocity and angular acceleration?

